National University of Computer and Emerging Sciences 0091

Islamabad Campus

Technology Management (MG2007)

Sessional-1 Exam

Section: A, B, and C

Degree program: BCS08

Date: February 24rd 2026

Course Instructor(s)

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Total Time (Hrs): 01

Total Marks: 30

Total Questions: 3

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Roll No **Course Section** **Student Signature**

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Attempt all the questions.

[CLO 1: Explain the strategic role of technology and innovation management in achieving organizational competitiveness

CLO 2: Analyze the frameworks and processes involved in planning, implementing, reviewing, and controlling innovation initiatives.]

Q1: This question consists of 10-MCQs. Mark your choice for each of the MCQ on the exam paper. **[20 marks]**

1. The central paradox discussed in the Innovator's Dilemma is that:

A) Firms fail because they don't invest in R&D

B) Firms fail because they ignore customers

C) Firms fail despite doing everything "right"

D) Firms fail due to lack of competition

2. The "mudslide hypothesis" suggested firms fail because:

A) They lack capital

B) Technology changes too fast

C) Customers shift preferences

D) Managers are incompetent

3. Christensen concluded the mudslide hypothesis was:

A) Fully correct

B) Partially correct

C) Wrong

D) Unrelated

4. In sustaining innovations within disk drives:

A) Entrants dominated

B) Incumbents consistently led

C) Government firms dominated
D) No firm succeeded

5. Value networks define:

A) Patent ownership
B) Stock valuation
C) What customers value and margins required
D) CEO compensation

6. Incumbents fail mainly because:

A) They ignore R&D
B) They lack engineers
C) They rationally focus on profitable customers
D) They avoid marketing

7. Entrants dominate disruptive innovation because:

A) They have more capital
B) They are free from legacy cost structures
C) Governments support them
D) They avoid competition

8. Thin-film and magneto-resistive heads are examples of:

A) Disruptive innovations
B) Marketing innovations
C) Sustaining innovations
D) Organizational change

9. The Innovator's Dilemma occurs because:

A) Firms lack funds
B) Firms choose sustaining over disruptive innovation
C) Firms ignore customers
D) Markets collapse

10. In portable value networks, customers value:

A) Maximum capacity only
B) Ruggedness and low power
C) Large size
D) High customization

[CLO 2: Analyze the frameworks and processes involved in planning, implementing, reviewing, and controlling innovation initiatives.]

Question 2: Case Scenario (10 Marks)

Between 1956 and 1995, the disk drive industry became a classic example of innovation-driven change. Early leaders like **IBM** dominated the 14-inch drive market, serving mainframe customers who valued high storage capacity and reliability. Performance improvements followed a typical **technology S-curve**: slow early progress, rapid improvement, and eventual maturity.

During this period, firms invested heavily in **sustaining innovations**, such as improvements in recording density and speed. These innovations helped companies move upward along the existing S-curve.

However, new architectural designs emerged:

- 8-inch drives (minicomputers)
- 5.25-inch drives (desktop PCs, introduced by **Seagate Technology**)
- 3.5-inch drives (portable computers, popularized by **Conner Peripherals**)

Initially, these smaller drives had lower capacity and worse performance compared to 14-inch drives. Mainframe customers were not interested. But these new drives started a **new S-curve**, targeting different performance attributes such as size, portability, and low power consumption.

Established firms continued improving the old technology (moving up the existing S-curve), while entrants focused on the new curve. Over time, the new S-curve improved rapidly and overtook the old one, leading to the failure of many established firms.

(a). Using the S-curve model, explain why established firms focused on improving 14-inch drives instead of investing in smaller drives. **(3 Marks)**

(b). Differentiate between sustaining and disruptive innovation using examples from the case. **(3 Marks)**

(c). Explain how value networks and technology patterns together caused established firms to fail. **(4 Marks)**

[CLO 1: Explain the strategic role of technology and innovation management in achieving organizational competitiveness

Question 3. What are the benefits and challenges of technology & innovation for the businesses, society, and international community? **(10 marks)**

Q.2

(a)

Established firms wanted to benefit from the high profit margins of their pre-existing consumers instead of focusing on new entrants/market which typically have low margins.

In context of S curve, the 14-inch drives were in their prime having passed the initial slow growth. Whereas, the smaller drives were at the start of the S curve with slow growth.

(b)

↳ Sustaining innovations improve the current product/process. They are incremental and benefit from the already mass adoption of the current product/process.
Example: IBM improving its 14 inch hard drive with greater capacity and faster speeds.

↳ ~~Disruptive innovations introduce a~~

↳ Disruptive Innovations introduce new products/processes altogether. They target new markets. Initially they are worse compared to the matured products/processes but eventually surpass them.
Example : Newer 3.5 inch drives by Conner Peripherals which have smaller form factor and used a new technology

1.5

(c)

↳ Value Networks are the context (users, product, society, regulations) within which companies operate in.
Now, established firms tend to fixate themselves on their existing value networks. This leads to firms only focusing on what the existing customer wants instead of innovating for new customers. Causing stagnant markets. Technology often exacerbates this issue.
Example : Nokia focused on their existing value network instead of innovating to capture the initial smartphone leading to their downfall. (next page)

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— Also since newer technology has made it easier for entrants, this allows for newer firms to rapidly multiple newer value networks whereas legacy firms do not explore.

Q.3

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QUESTION #03

We will discuss benefits and challenges one-by-one.

1. BUSINESSES :-

Benefits :-

- Business can reduce cost of redundant or reoccuring tasks by Automation.
- Business can now track customer need more easily with advent of internet.
- Rapid prototyping is now possible with state of the art AI Tools.

Challenges :-

- Dominos Effect. Adopting new technology can cause whole organizational structure to transform.
- Complexity of architecture with every new technology.

2. SOCIETY :-

Benefits :-

- People can now see prices of product easily. In past same product was sold differently.

- Easy to connect with distant people.
- Easy on-door delivery of everything available on internet.

Challenges:-

- lesser face-to-face interaction causing less ~~set~~ socializing.
- Ethical concerns over personal identity.

3) INTERNATIONAL COMMUNITY:-

Benefits :-

- Good salary package for people of Developing countries working remote.
- Easy to connect with cross-functional and cross-country teams

Challenges :-

- Difficult to maintain cross-country teams due to time-zones.
- local jobs compromised in developed countries as developing are offering cheap and quality work.